

INDIA TECH CAREER INTELLIGENCE DASHBOARD

A Data Analytics & Power BI Project on the Indian Technology Job Market

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Tools: Python | Pandas | Power BI | DAX | GitHub

Dataset: India Tech Job Market (2024-2026) | 5,000 Listings | 17 Attributes | 109 Skills

1. Project Overview

The Indian technology job market is evolving rapidly, making it difficult for students, job seekers, and career changers to understand which skills are most in demand, which roles offer the highest salaries, and where the best opportunities exist.

This project analyzes a large dataset of Indian technology job postings and presents interactive insights through a Power BI dashboard. The dashboard enables users to explore hiring trends, salary patterns, required skills, and career opportunities using dynamic filters and visualizations.

2. Problem Statement

Many aspiring professionals struggle to answer questions such as:

- Which technology roles have the highest demand?
- Which skills should I learn?
- Which cities offer the best career opportunities?
- Which job roles provide higher salaries?
- What qualifications are commonly required?
- Which companies hire the most?

Existing job portals provide thousands of listings but do not summarize market trends in a meaningful way. This project transforms raw job data into actionable career insights.

3. Objectives

The primary objectives of this project are:

- Analyze hiring trends in the Indian technology sector.
- Identify the highest paying technology roles.
- Discover the most in-demand technical skills.
- Compare salary trends across cities and experience levels.
- Build an interactive dashboard for career exploration.
- Help users make data-driven career decisions.

4. Dataset Information

Dataset Name: India Tech Job Market (2024-2026)

5,000	17	109
Job Listings	Attributes	Unique Skills

Key Features:

Job Title	Company	Company Type	City
Experience Level	Salary (LPA)	Skills Required	Education Required
Work Mode	Openings	Applicants	Company Rating

5. Tools & Technologies

- Python
- Pandas
- Jupyter Notebook
- Power BI
- DAX
- GitHub

6. Project Workflow

- 1 Data Collection
- 2 Data Understanding
- 3 Data Cleaning
- 4 Exploratory Data Analysis
- 5 Feature Engineering
- 6 Power BI Dashboard Development
- 7 Business Insights

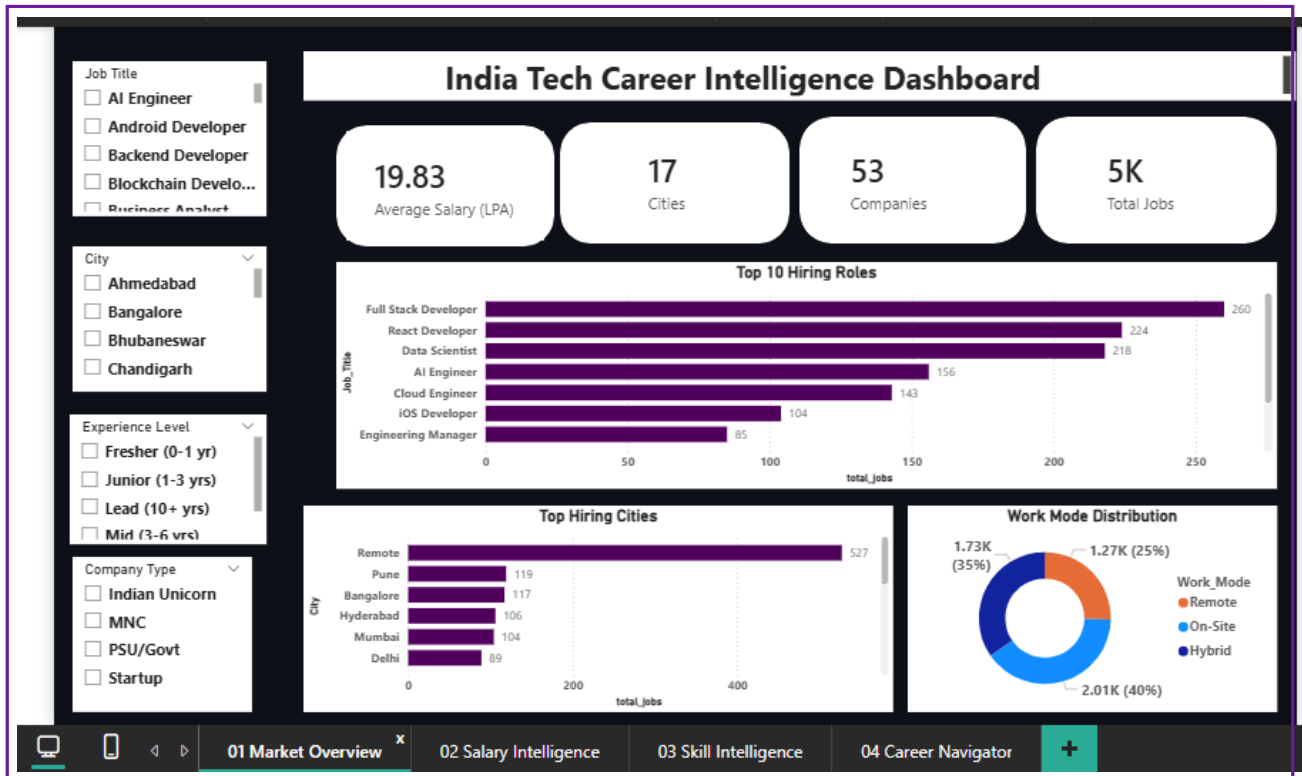
7. Dashboard Pages

The Power BI report is organized into four interconnected pages, each answering a distinct set of career-market questions. Every page carries slicers for Job Title, City, Experience Level, and Company Type, so the same story can be explored at the market level or narrowed down to a single role.

PAGE 1

Market Overview

Provide a complete overview of the Indian technology job market.



Screenshot: live Power BI dashboard view

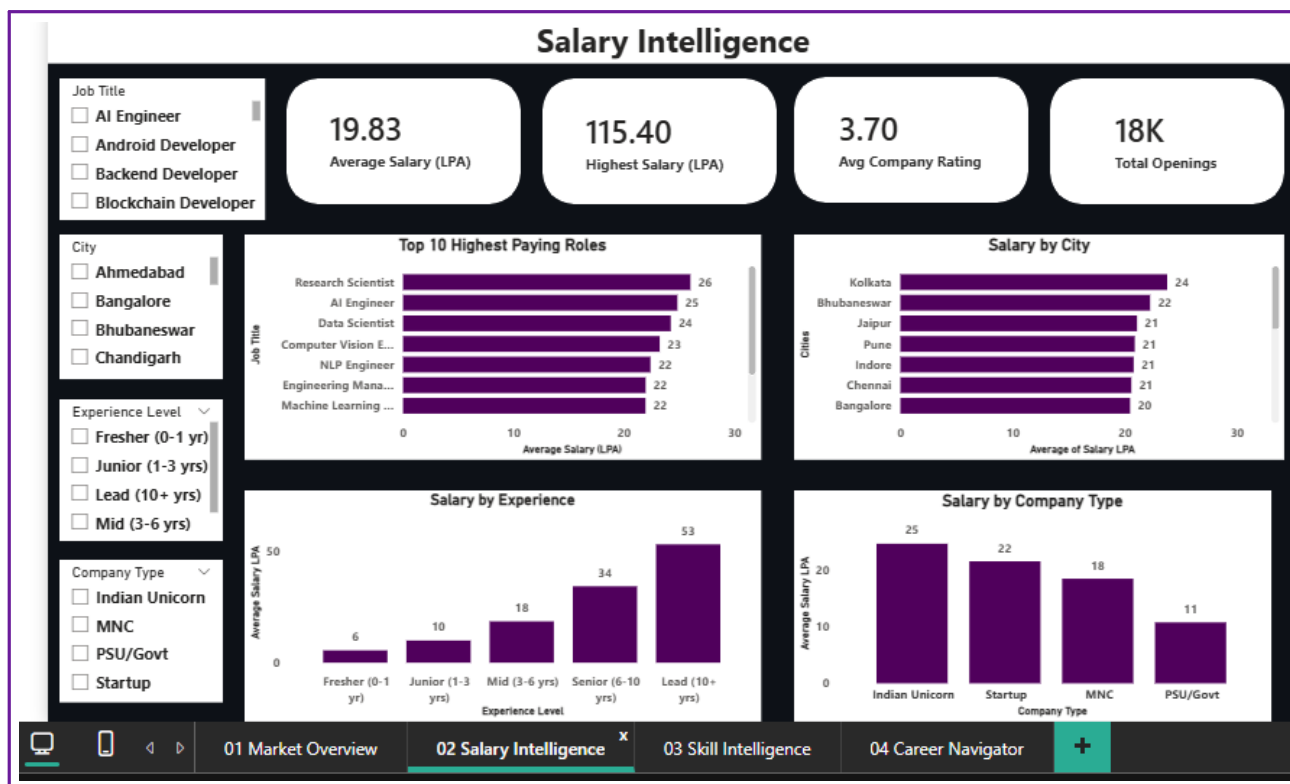
Includes:

- Total Jobs
- Total Companies
- Total Cities
- Average Salary
- Top Hiring Roles
- Top Hiring Cities
- Work Mode Distribution

Read on this page: Full Stack Developer and React Developer top the hiring charts at 260 and 224 job postings. Remote listings (35%) lead the work-mode split, followed closely by Hybrid (40%) and On-Site (25%), and Pune ranks as the second-highest hiring city after fully remote roles.

Salary Intelligence

Analyze salary trends across different factors.



Screenshot: live Power BI dashboard view

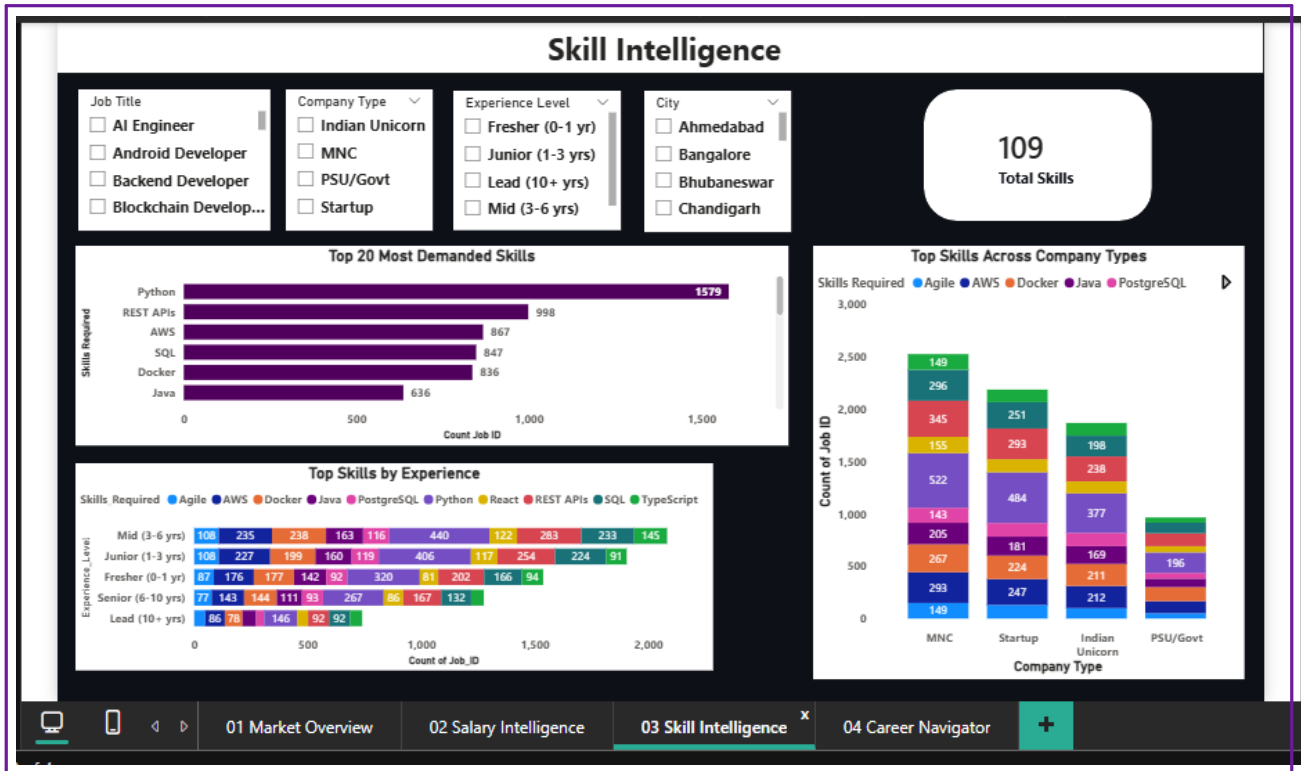
Includes:

- Highest Salary
- Average Salary
- Salary by Role
- Salary by Experience
- Salary by Company Type
- Salary by City

Read on this page: Research Scientist and AI Engineer command the highest average pay among all roles. Salary rises sharply with experience, from 6 LPA at Fresher level to 53 LPA at Lead level, and Indian Unicorns out-pay MNCs, Startups, and PSU/Govt employers on average.

Skill Intelligence

Identify the most valuable skills in the market.



Screenshot: live Power BI dashboard view

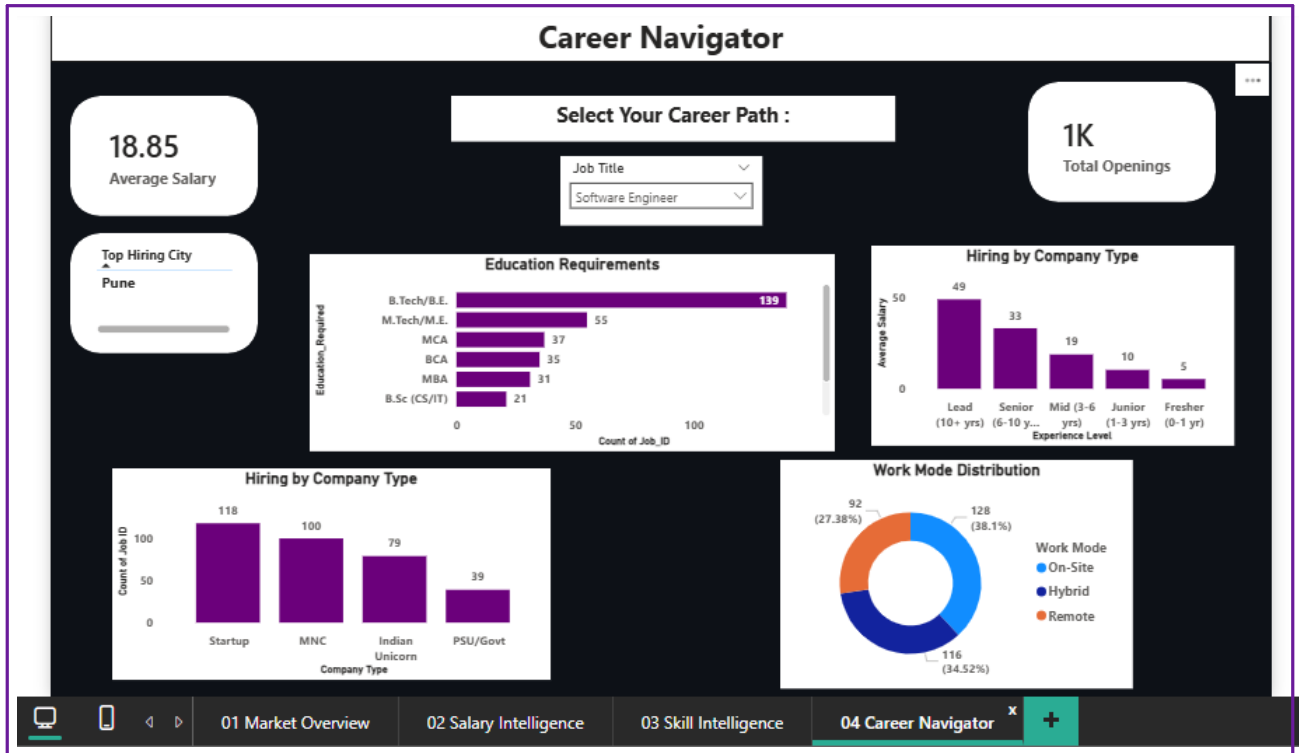
Includes:

- Most Demanded Skills
- Skills by Experience
- Skills Across Company Types

Read on this page: Python is the single most requested skill by a wide margin (1,579 mentions), ahead of REST APIs, AWS, SQL, Docker, and Java. Skill demand shifts with seniority: foundational languages dominate at Fresher level while Docker and cloud tooling gain weight at Mid and Senior levels.

Career Navigator

Allow users to explore career-specific insights by selecting a job role.



Screenshot: live Power BI dashboard view

Includes:

- Average Salary
- Total Openings
- Top Hiring City
- Education Requirements
- Company Hiring Pattern
- Work Mode Distribution

Read on this page: For a selected role such as Software Engineer, B.Tech/B.E. is the dominant qualification (139 postings), Pune is the top hiring city, Lead-level positions pay the highest average salary, and Startups post the most openings, ahead of MNCs, Indian Unicorns, and PSU/Govt.

8. Key Insights

- Software Engineer is among the most frequently advertised job roles.
- Python is the most demanded technical skill.
- Hybrid and On-Site work models dominate hiring.
- Salary increases significantly with experience level.
- Indian Unicorn companies generally offer higher salaries than MNCs and Startups.
- Different roles require different skill combinations and educational qualifications.

9. Business Impact

Students	Identify the right skills to learn; explore career paths; understand salary expectations.
Job Seekers	Compare job opportunities; discover high-demand roles; make informed career decisions.
Training Institutes	Design courses based on industry demand.
HR & Recruiters	Understand hiring trends; benchmark salaries; analyze demand across technologies.

10. Future Improvements

- Connect to real-time job APIs.
- Add monthly hiring trend analysis.
- Include salary prediction using Machine Learning.
- Add company-wise comparisons.
- Build a web-based dashboard.

11. Conclusion

This project demonstrates how Python and Power BI can transform raw job market data into meaningful insights. By combining data cleaning, exploratory analysis, and interactive visualization, the dashboard enables users to understand hiring trends, salary patterns, and skill demand in the Indian technology industry. It serves as a practical decision-support tool for students, professionals, recruiters, and educational institutions.